LECTURE-2

Chemistry

INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY
DELHI

2

The Chemical Context of Life

KEY CONCEPTS

2.1 Matter consists of chemical elements in pure form and in combinations called compounds

2.2 An element's properties depend on the structure of its atoms

2.3 The formation and function of molecules depend on chemical bonding between atoms

2.4 Chemical reactions make and break chemical bonds

General ideas

- Living organisms and the world they live in are subject to the basic **laws of physics and chemistry**.

- Biology is a **multidisciplinary** science, drawing on insights from other sciences.

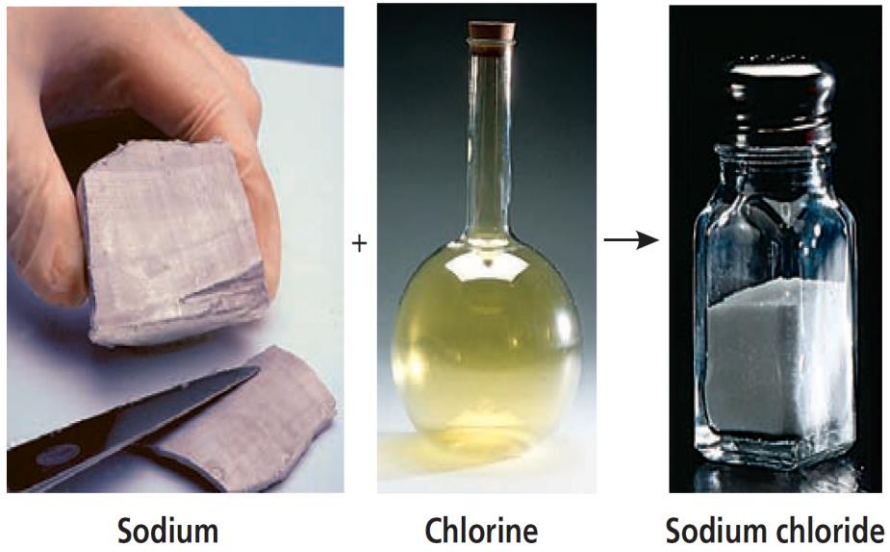
- Life can be organized into a **hierarchy of structural levels**.

  - *Atoms are organized into molecules, and molecules are organized into cells.*
  - *At each successive level, additional **emergent properties appear**.*
  - *Somewhere in the transition from molecules to cells, we cross **the boundary between nonlife and life**.*

Matter, elements, and compounds

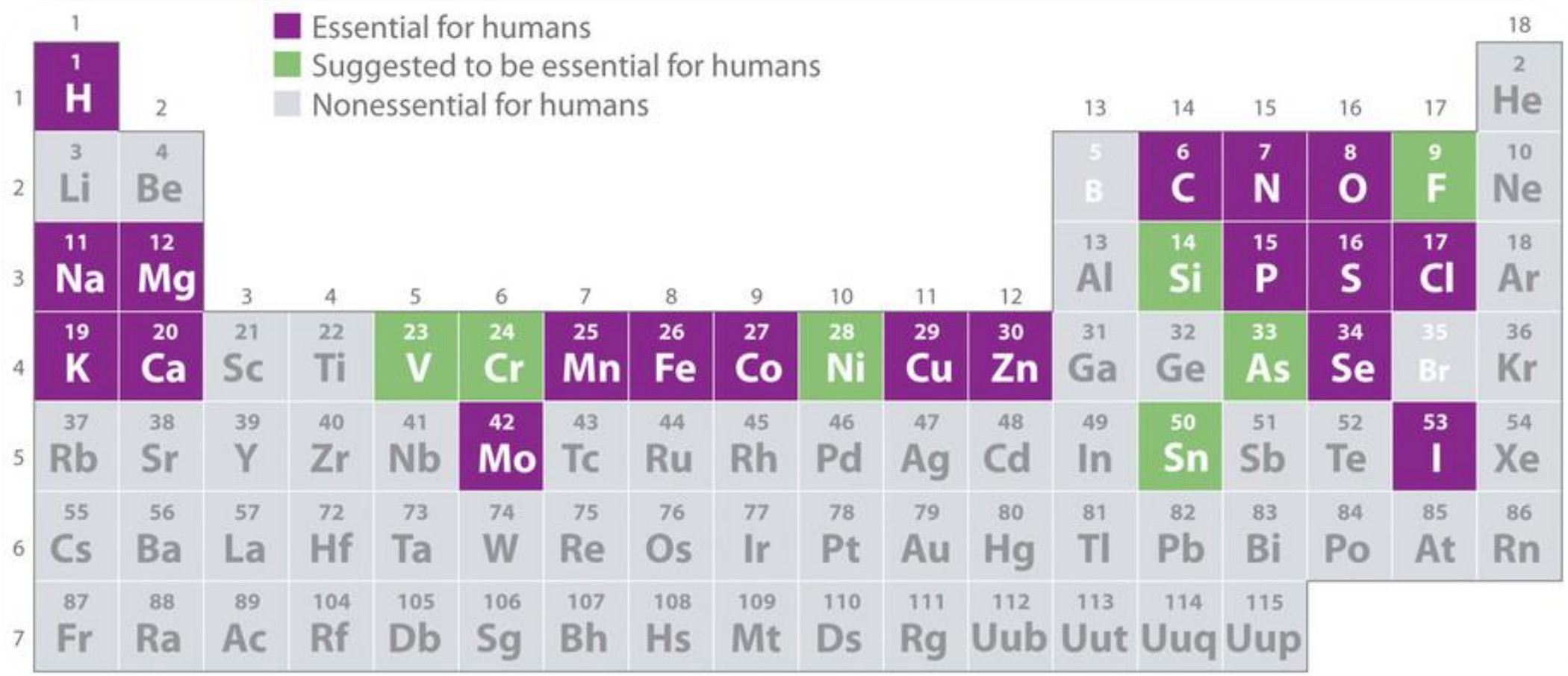
- Organisms are composed of **matter**, defined as ‘anything that takes up space and has mass’.
  - Matter is made up of elements.
- An **element** is a substance that cannot be broken down into other substances by chemical reactions.
  - *There are 92 naturally occurring elements.*
  - *Each element has a unique symbol, usually the first one or two letters of its name. Some symbols are derived from Latin or German names.*
- A **compound** is a substance that consists of two or more elements in a fixed ratio.

Emergent property



▲ **Figure 2.2 The emergent properties of a compound.** The metal sodium combines with the poisonous gas chlorine, forming the edible compound sodium chloride, or table salt.

Periodic table: Elements of life



Essential elements

- ***Twenty-five chemical elements are essential for life.***

- About 20–25% of the 92 naturally occurring elements are **essential elements** that are necessary for living things.
  - ✓ There is some variation: **humans have 25 essential elements**, while plants have only 17.

- **Four elements—oxygen (O), carbon (C), hydrogen (H), and nitrogen (N)—make up 96% of living matter (COHN).**
  - ✓ Most of the remaining 4% of an organism's weight consists of *calcium (Ca), phosphorus (P), potassium (K), and sulfur (S)*.

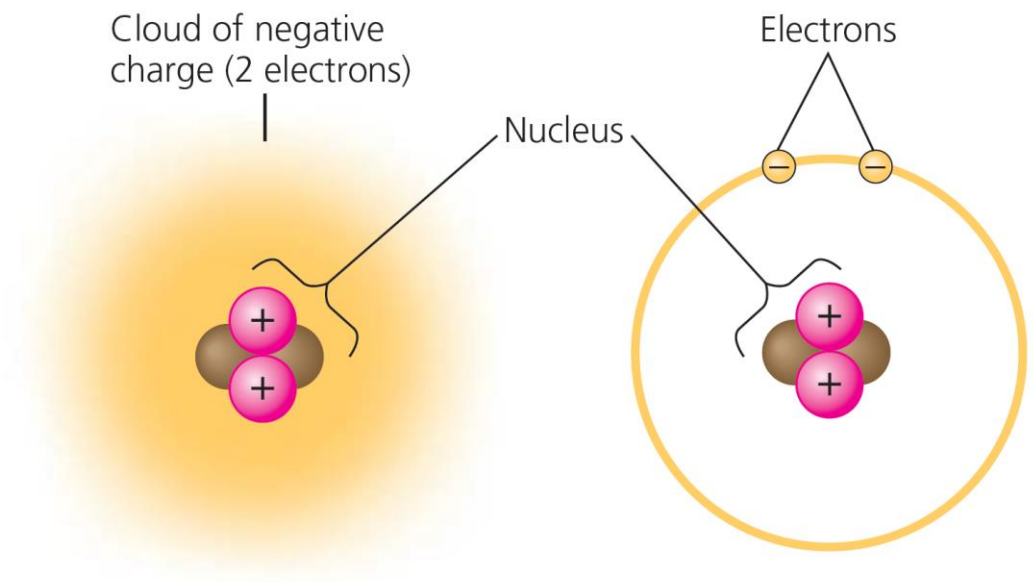
Table 2.1 Elements in the Human Body

Element	Symbol	Percentage of Body Mass (including water)	
Oxygen	O	65.0%	96.3%
Carbon	C	18.5%	
Hydrogen	H	9.5%	
Nitrogen	N	3.3%	
Calcium	Ca	1.5%	3.7%
Phosphorus	P	1.0%	
Potassium	K	0.4%	
Sulfur	S	0.3%	
Sodium	Na	0.2%	
Chlorine	Cl	0.2%	
Magnesium	Mg	0.1%	

Trace elements (less than 0.01% of mass): Boron (B), chromium (Cr), cobalt (Co), copper (Cu), fluorine (F), iodine (I), iron (Fe), manganese (Mn), molybdenum (Mo), selenium (Se), silicon (Si), tin (Sn), vanadium (V), zinc (Zn)

Three important particles in the atom

- An **atom** is the smallest unit of matter that still retains the properties of an element.
- **Neutrons**, **protons**, and **electrons** are the three important particles in an atom.
- Protons and electrons are electrically charged. Each proton has one unit of positive charge, and each electron has one unit of negative charge. A neutron, as its name implies, is electrically neutral.
- Protons and neutrons are packed together tightly in a dense core, or **atomic nucleus**, at the center of an atom; protons give the nucleus a positive charge.
- The rapidly moving electrons form a "cloud" of negative charge around the nucleus, and it is the attraction between opposite charges that keeps the electrons in the vicinity of the nucleus.



(a) This model represents the two electrons as a cloud of negative charge.

(b) In this more simplified model, the electrons are shown as two small yellow spheres on a circle around the nucleus.

▲ **Figure 2.4 Simplified models of a helium (He) atom.** The helium nucleus consists of 2 neutrons (brown) and 2 protons (pink). Two electrons (yellow) exist outside the nucleus. These models are not to scale; they greatly overestimate the size of the nucleus in relation to the electron cloud.

Atom and subatomic particles

- A neutron and a proton have almost identical mass: about 1.7×10^{-24} gram per particle.
- Grams and other conventional units are not very useful for describing the mass of objects that are so minuscule.
- Thus, for atoms and subatomic particles (and for molecules, too), we use a unit of measurement called the Dalton.
- **Dalton:** The mass of a neutron or a proton is close to 1 dalton.
- The mass of an electron is about 1/2,000 the mass of a neutron or proton.
- This number of protons is the element's unique **atomic number**.
- The **mass number** is the sum of the number of protons and the number of neutrons in the nucleus of an atom.

Mass number = number of protons + neutrons
= 23 for sodium

$^{23}_{11}Na$

Atomic number = number of protons
= number of electrons in a neutral atom
= 11 for sodium

Number of neutrons = mass number − atomic number
= 23 − 11 = 12 for sodium

Carbon isotopes

- Let's consider the three naturally occurring isotopes of the element carbon, which has the atomic number 6.
- The most common isotope is carbon-12, ${}_{6}^{12}C$, which accounts for about 99% of the carbon in nature.
- The isotope ${}_{6}^{12}C$ has 6 neutrons.
- Most of the remaining 1% of carbon consists of atoms of the isotope, ${}_{6}^{13}C$ with 7 neutrons.
- Third, even rarer isotope, ${}_{6}^{14}C$, has 8 neutrons.
- Notice that all three isotopes of carbon have 6 protons; otherwise, they would not be carbon.
- Although the isotopes of an element have slightly different masses, they behave identically in chemical reactions
- Both ${}_{12}C$ and ${}_{13}C$ are stable isotopes, meaning that their nuclei do not have a tendency to lose subatomic particles, a process called decay.
- The isotope ${}_{14}C$, however, is unstable, or radioactive.
- A **radioactive isotope** is one in which the nucleus decays spontaneously, giving off particles and energy.
- When the radioactive decay leads to a change in the number of protons, it transforms the atom to an atom of a different element.

8) Carbon-14 has the same _____.

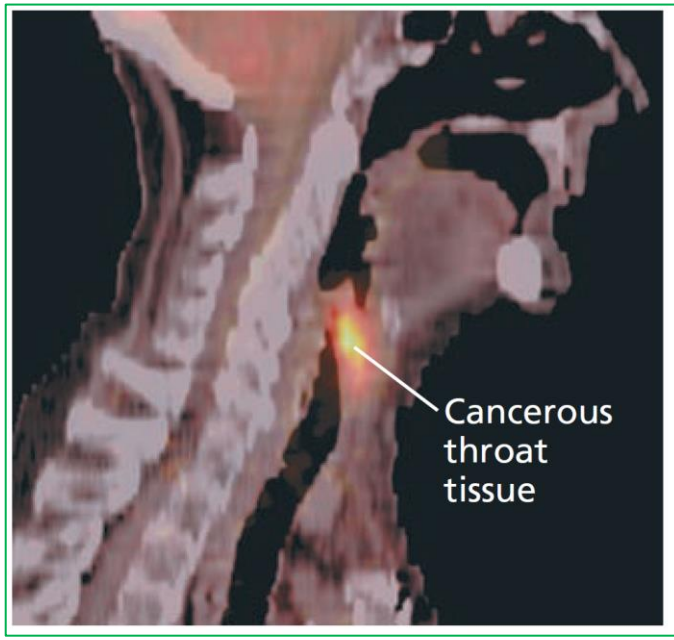
A) atomic number and atomic mass as carbon-12
B) atomic number and thus number of neutrons as carbon-13
C) atomic mass as both carbon-12 and carbon-13
D) number of protons but more neutrons than carbon-12

Answer: D

Isotopes and its use in research

- Two atoms of the same element that **differ in the number of neutrons** are called **isotopes.**
- In nature, an element occurs as a mixture of isotopes.
- The nuclei of **some isotopes are unstable** and decay spontaneously, emitting particles and energy.
- ^{14}C is one of these unstable isotopes, or radioactive isotopes.
- When ^{14}C decays, one of its neutrons is converted to a proton and an electron. In this process, ^{14}C is converted to ^{14}N, a different element.

- Radioactive isotopes have many **applications in biological research**.
  - ✓ Radioactive decay rates can be used **to date fossils**.
  - ✓ Radioactive isotopes can be used to **trace atoms through metabolic processes**.
- Radioactive isotopes are also used to **diagnose medical disorders**.

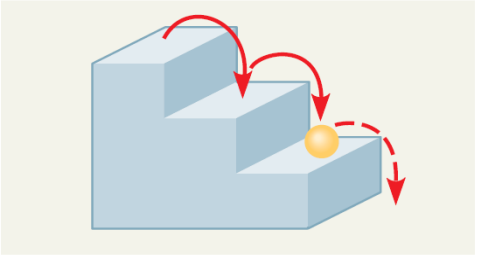


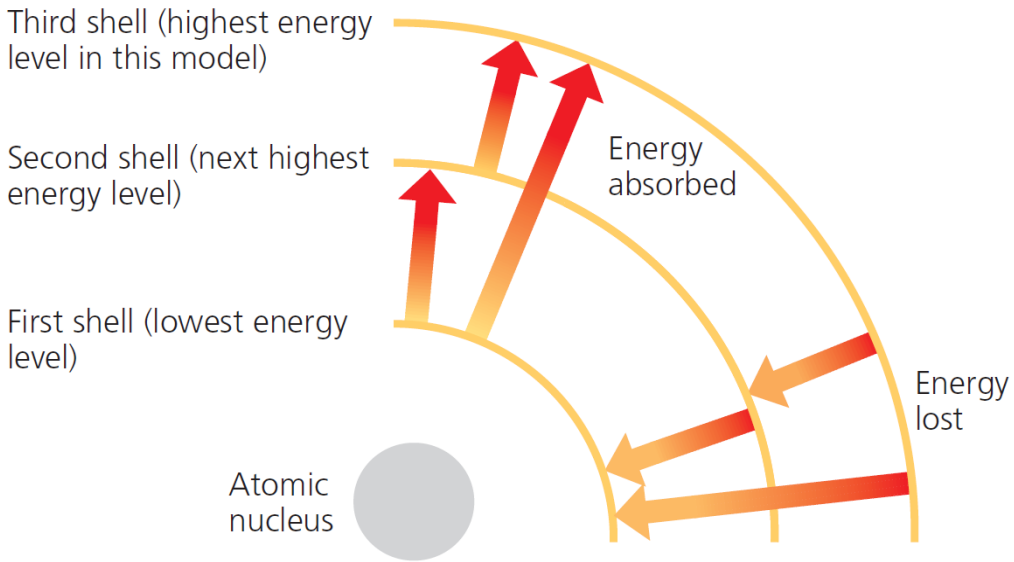
◀ **Figure 2.5 A PET scan, a medical use for radioactive isotopes.** PET, an acronym for positron-emission tomography, detects locations of intense chemical activity in the body. The bright yellow spot marks an area with an elevated level of radioactively labeled glucose, which in turn indicates high metabolic activity, a hallmark of cancerous tissue.

Energy levels of electrons

- **Potential energy** is the energy that matter stores because of its structure or location.
- **The chemical behavior of an atom is determined by its electron configuration**—the distribution of electrons in its electron shells.
- **Changes in** an electron's potential **energy can occur only in steps** of a fixed amount, moving the electron to a fixed location relative to the nucleus.
- The different states of potential energy of the electrons of an atom are called **electron shells**.

(a) A ball bouncing down a flight of stairs provides an analogy for energy levels of electrons, because the ball can come to rest only on each step, not between steps.



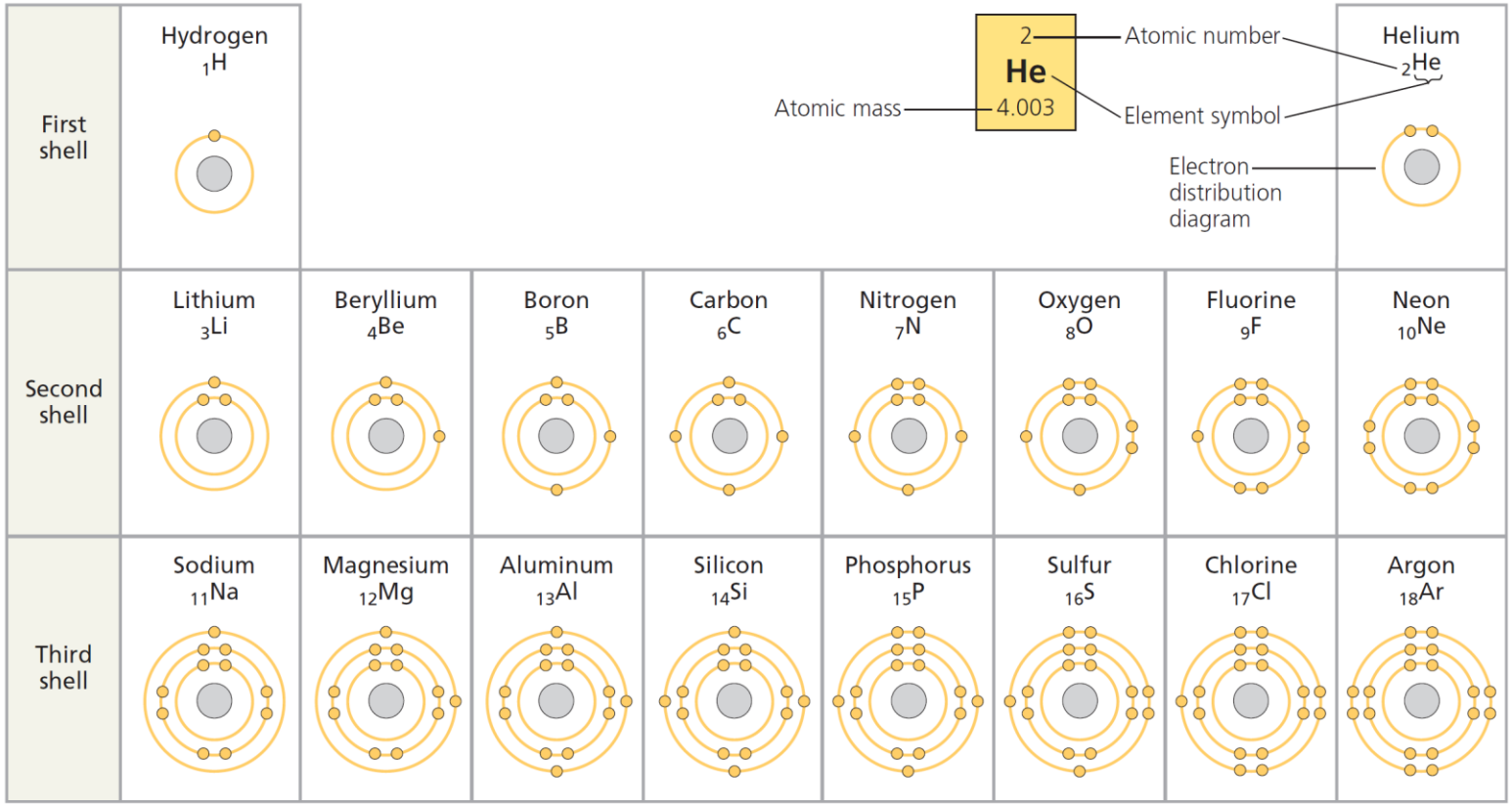


(b) An electron can move from one shell to another only if the energy it gains or loses is exactly equal to the difference in energy between the energy levels of the two shells. Arrows in this model indicate some of the stepwise changes in potential energy that are possible.

▲ **Figure 2.6 Energy levels of an atom's electrons.** Electrons exist only at fixed levels of potential energy called electron shells.

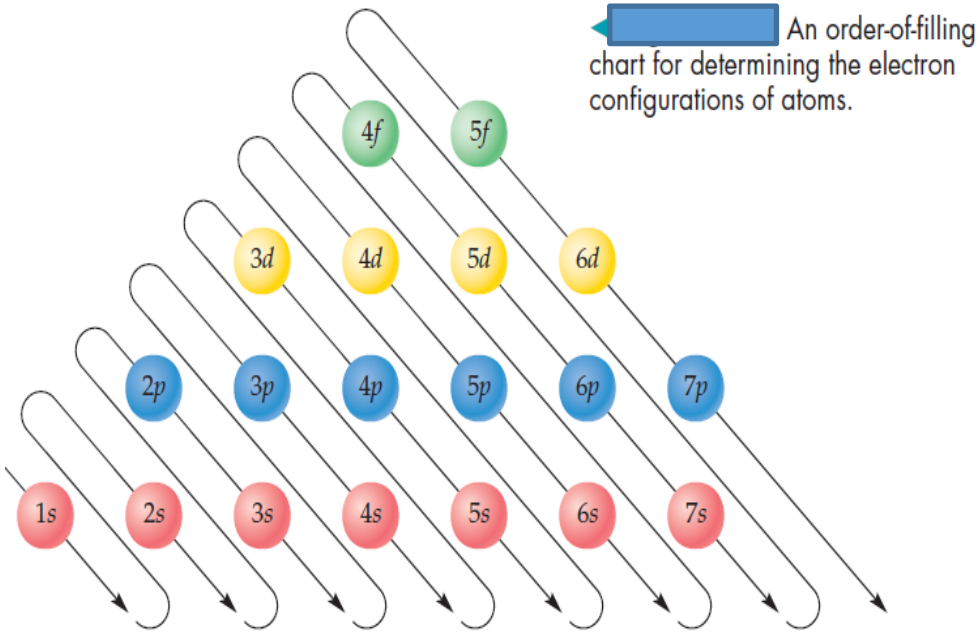
Energy levels of electron

❖ The chemical behavior of an atom depends mostly on the number of electrons in its outermost shell, the **valence shell.**



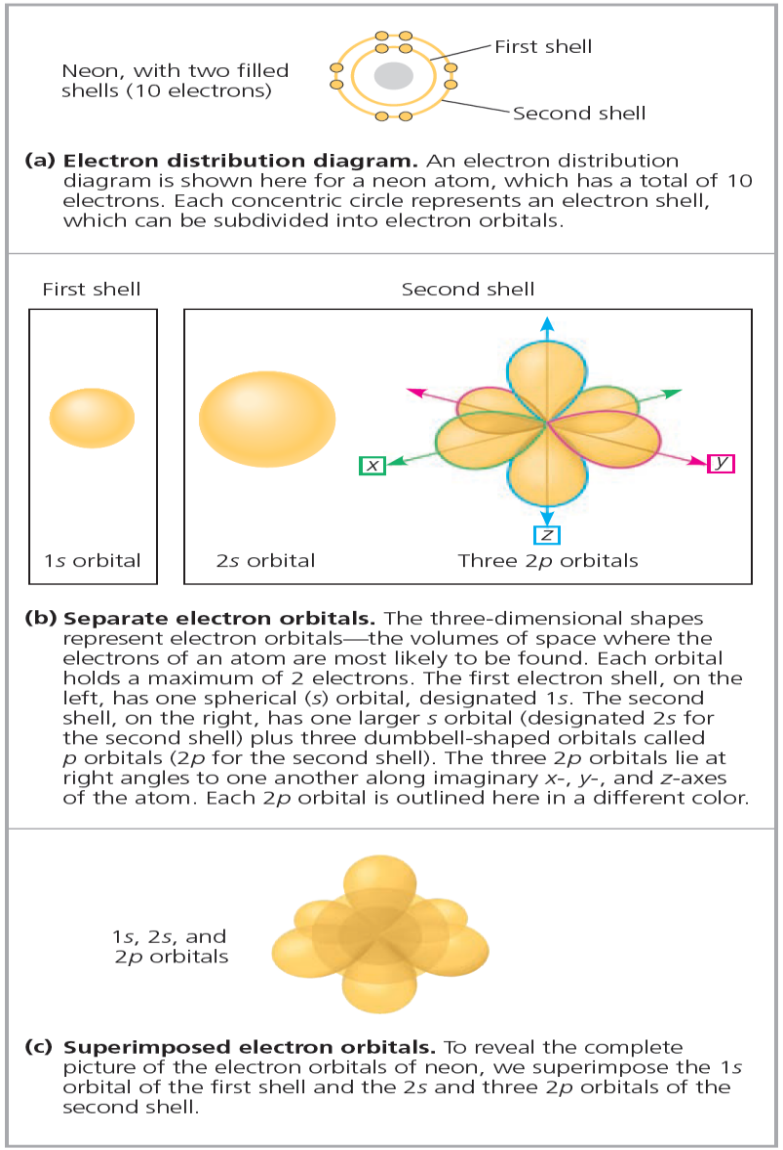
▲ **Figure 2.7 Electron distribution diagrams for the first 18 elements in the periodic table.** In a standard periodic table (see Appendix B), information for each element is presented as shown for helium in the inset. In the diagrams in this table, electrons are represented as yellow dots and electron shells as concentric circles. These diagrams are a convenient way to picture the distribution of an atom's electrons among its electron shells, but these simplified models do not accurately represent the shape of the atom or the location of its electrons. The elements are arranged in rows, each representing the filling of an electron shell. As electrons are added, they occupy the lowest available shell.

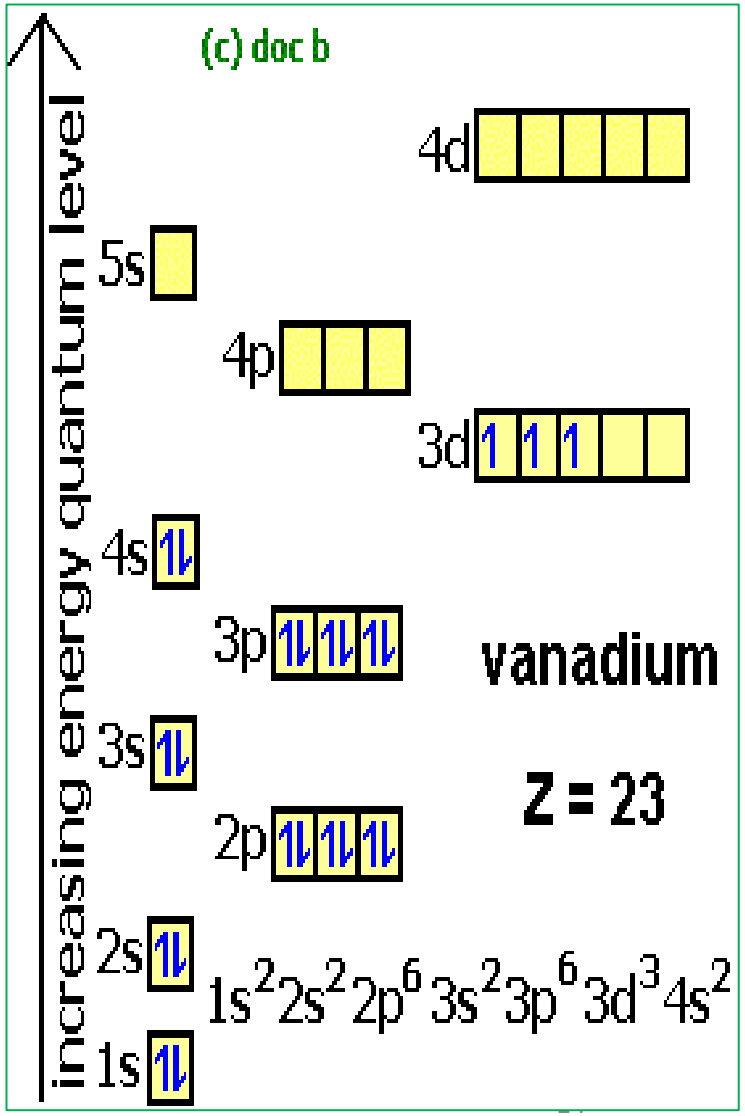
? *What is the atomic number of magnesium? How many protons and electrons does it have? How many electron shells? How many valence electrons?*



Electronic orbitals and arrangement of electrons

- It is convenient to use two-dimensional concentric-circle diagrams to represent the spatial occupation of electrons around the protons.
- These diagrams symbolize three-dimensional electron shells, each concentric circle represents only the *average* distance between an electron in that shell and the nucleus.
- The three-dimensional space where an electron is found 90% of the time is called an **orbital**.
- The reactivity of atoms arises from the presence of unpaired electrons in one or more orbitals of their valence shells.

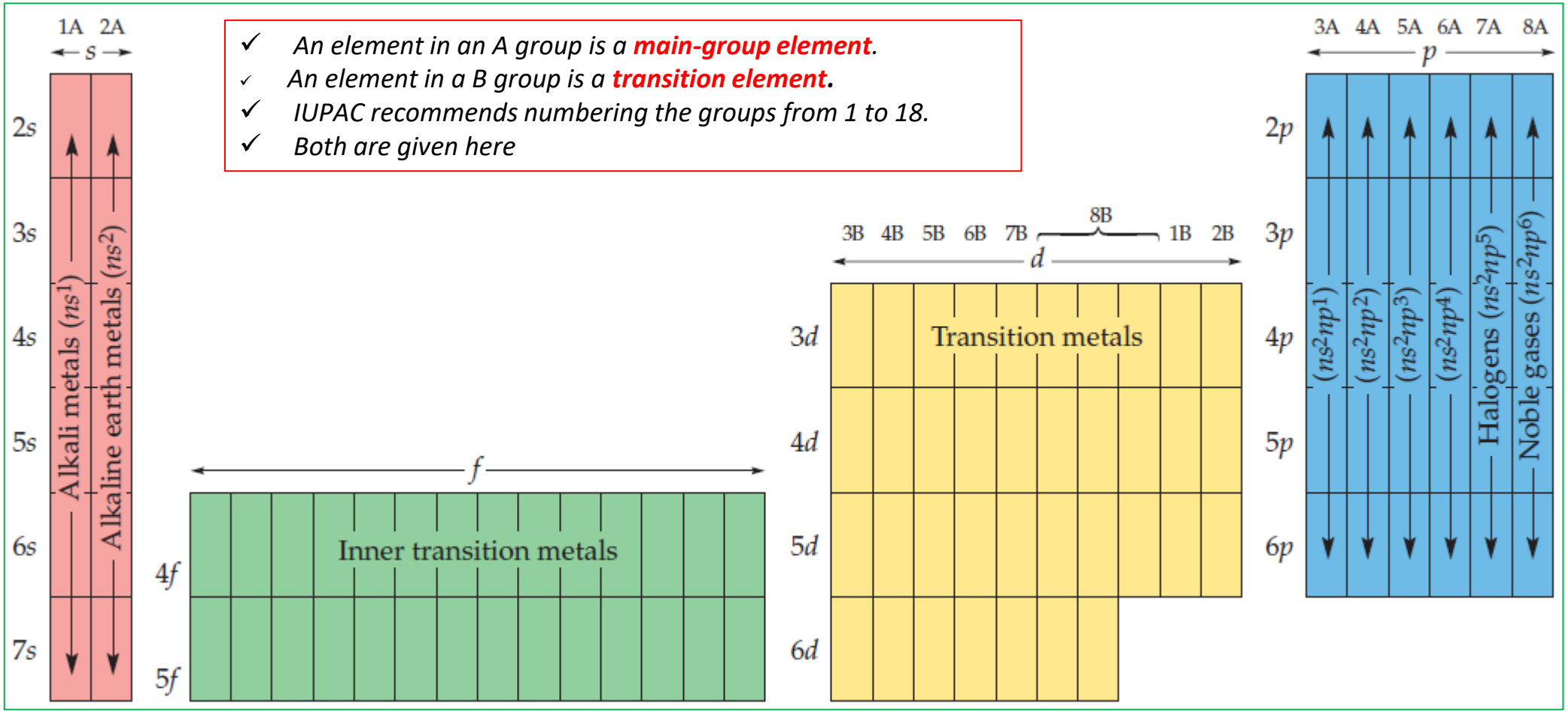




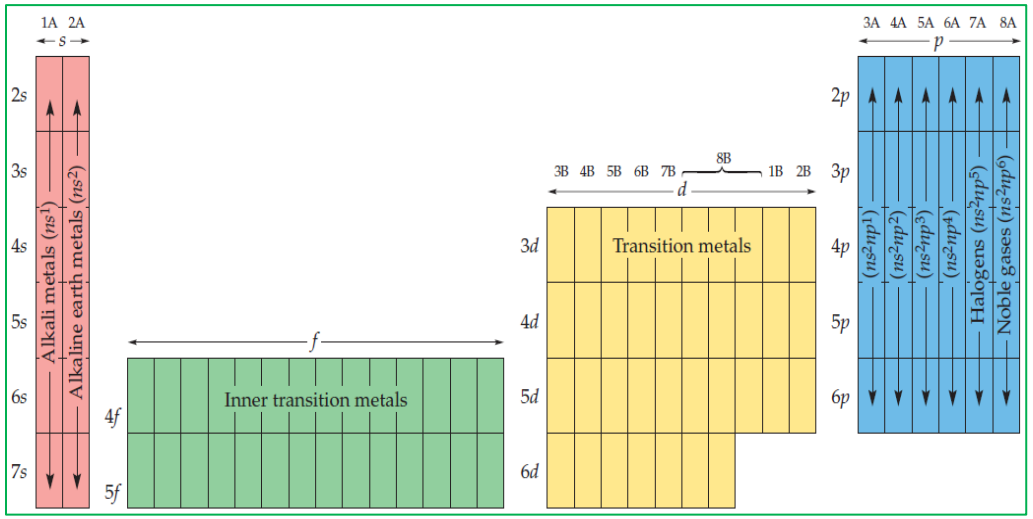
Importance of periodic table

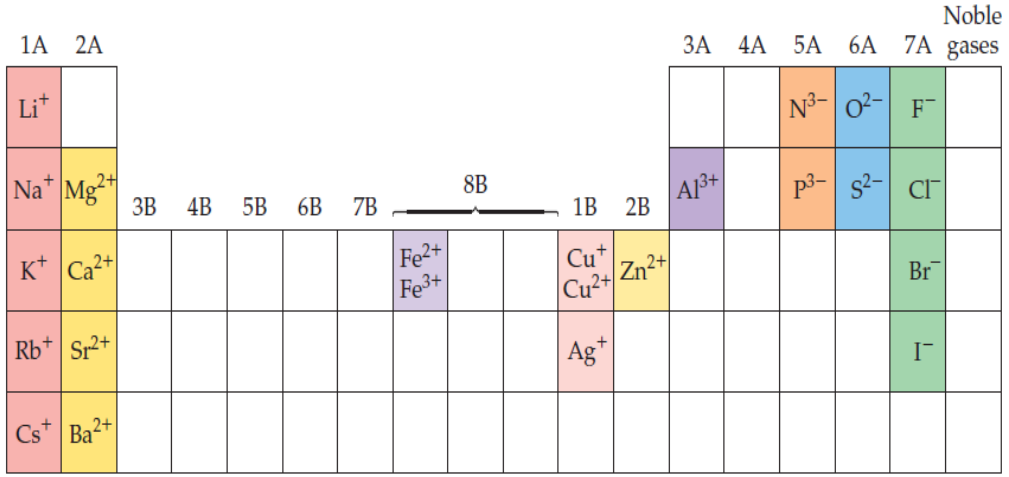
In periodic table, each vertical column is a **group** or *family*. Elements in a group have similar chemical properties.

A horizontal row of the periodic table is called a **period**. The properties of elements change in a recurring manner across a period.



Periodic relationships of simple ions





▲ **Figure 4.4** Periodic relationships of some simple ions. Many of the transition elements (B groups) can form different ions with different charges.

- Each atom of a metal tends to give up electrons from its outer shell, and each atom of a nonmetal tends to take on enough electrons to *complete, or close, its valence shell* (s^2p^6).
- In other words, atoms are driven toward becoming ions to have eight—an *octet* of—electrons in their valence shells.
- An octet of electrons is characteristic of all noble gases except helium.

Table 4.2 Symbols and Names for Some Simple (Monatomic) Ions

Group	Element	Name of Ion	Symbol for Ion
1A	Hydrogen	Hydrogen ion	H^+
	Lithium	Lithium ion	Li^+
	Sodium	Sodium ion	Na^+
	Potassium	Potassium ion	K^+
2A	Magnesium	Magnesium ion	Mg^{2+}
	Calcium	Calcium ion	Ca^{2+}
3A	Aluminum	Aluminum ion	Al^{3+}
5A	Nitrogen	Nitride ion	N^{3-}
	Phosphorus	Phosphide ion	P^{3-}
6A	Oxygen	Oxide ion	O^{2-}
	Sulfur	Sulfide ion	S^{2-}
7A	Fluorine	Fluoride ion	F^-
	Chlorine	Chloride ion	Cl^-
	Bromine	Bromide ion	Br^-
	Iodine	Iodide ion	I^-
1B	Copper	Copper(I) ion (cuprous ion)	Cu^+
		Copper(II) ion (cupric ion)	Cu^{2+}
	Silver	Silver ion	Ag^+
2B	Zinc	Zinc ion	Zn^{2+}
8B	Iron	Iron(II) ion (ferrous ion)	Fe^{2+}
		Iron(III) ion (ferric ion)	Fe^{3+}

Q: What is the formula of the compound formed by the reaction of sodium and sulfur?

Sodium is in group 1A; the sodium atom has one valence electron. Sulfur is in group 6A; the sulfur atom has six valence electrons.

$$Na\cdot \qquad \cdot\ddot{\underset{\cdot\cdot}{S}}\cdot$$

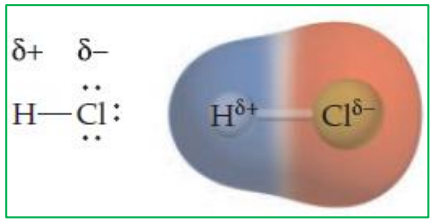
Sulfur needs two electrons to gain an argon configuration, but sodium has only one to give. The sulfur atom therefore must react with two sodium atoms.

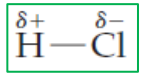
$$\begin{matrix} Na\cdot \\ Na\cdot \end{matrix} + \cdot\ddot{\underset{\cdot\cdot}{S}}\cdot \longrightarrow \begin{matrix} Na^+ \\ Na^+ \end{matrix} + :\ddot{\underset{\cdot\cdot}{S}}:^{2-}$$

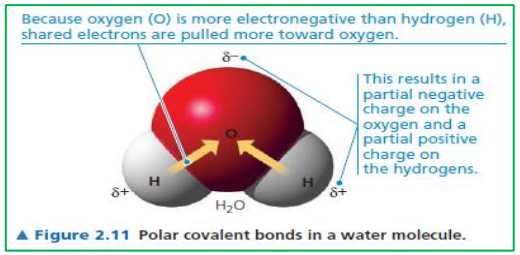
The formula of the compound, called sodium sulfide, is Na_2S.

Electronegativity

- The **electronegativity** of an element can be viewed as its ability to attract electrons *in a molecule*.
- The more electronegative an atom is, the greater its tendency to pull the electrons toward its end of the covalent bond.
- The atoms to the right in the periodic table are, in general, more electronegative than those to the left.
- The American chemist Linus Pauling (1901–1994) devised a scale of relative electronegativity values by assigning fluorine, the most electronegative element, a value of 4.0.







▲ **Figure 2.11** Polar covalent bonds in a water molecule.

- Here, chlorine is more electronegative. As a result, the shared electron pair will be closer to chlorine, making the chlorine end of the bond more negative compared to the hydrogen end.
- An unequal sharing of an electron pair (or pairs) leads to bond polarity, and the bond is called a **polar covalent bond**.
- By comparison, the bonds where the electron pairs are shared equally, such as those you have already seen in the diatomic molecules of hydrogen, chlorine, oxygen, and nitrogen, are **nonpolar covalent bonds.**

Octet config.

Electropositive

Electronegative

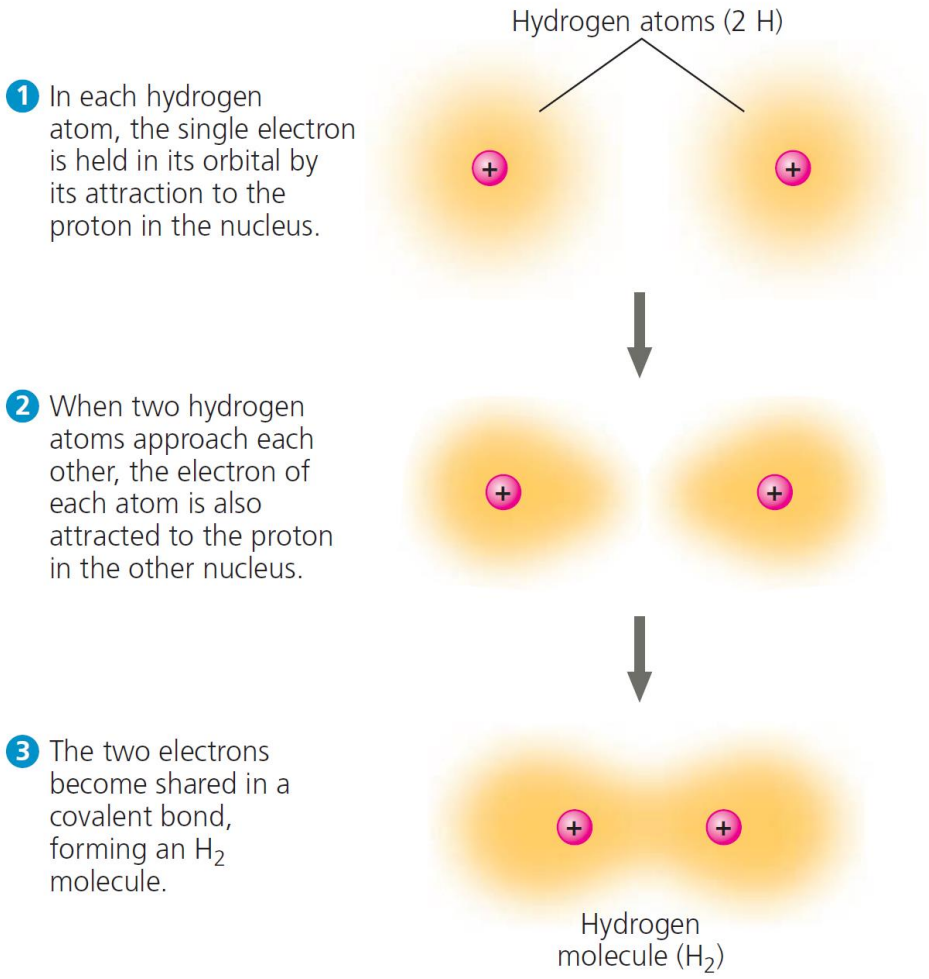
1A	2A	3B	4B	5B	6B	7B	8B			1B	2B	3A	4A	5A	6A	7A	Noble gases
H 2.1																	
Li 1.0	Be 1.5											B 2.0	C 2.5	N 3.0	O 3.5	F 4.0	
Na 0.9	Mg 1.2											Al 1.5	Si 1.8	P 2.1	S 2.5	Cl 3.0	
K 0.8	Ca 1.0													As 2.0	Se 2.4	Br 2.8	
																I 2.5	

▲ Pauling electronegativity values are given for several common elements.

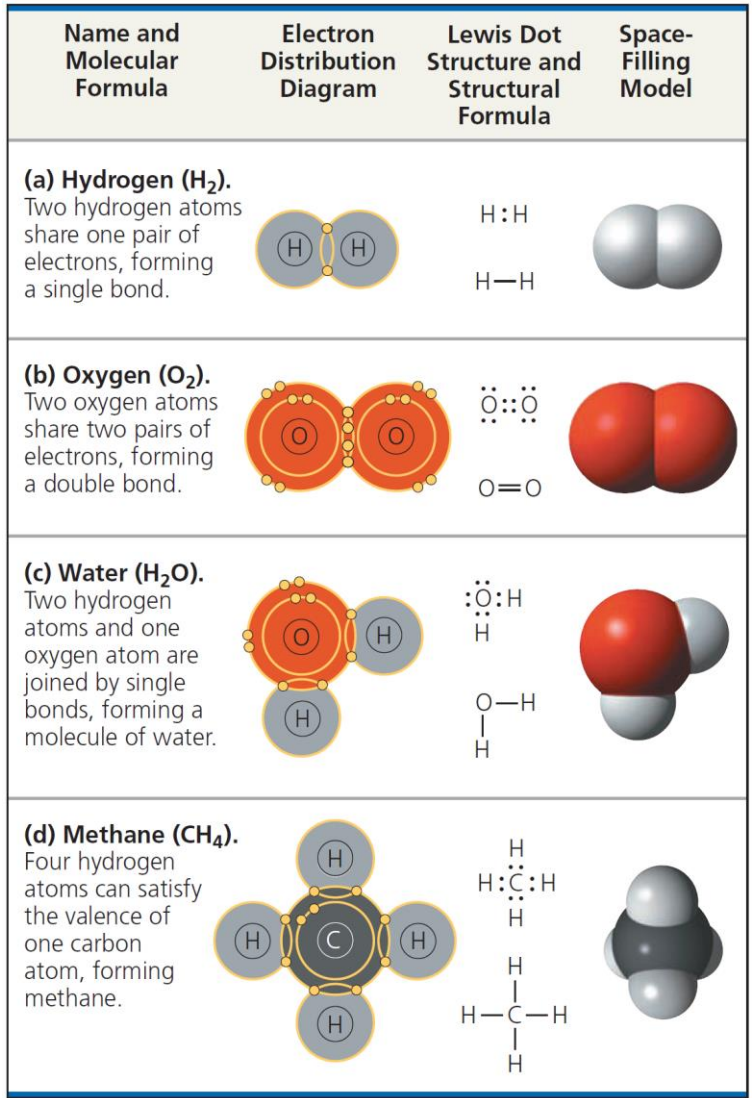
Electronegativity Difference	Bonding Type
< 0.5	Nonpolar covalent
0.5–2.0	Polar covalent
> 2.0	Ionic

Covalent bonds

❖ A **covalent bond** is the sharing of a pair of valence electrons by two atoms.



▲ **Figure 2.9 Formation of a covalent bond.**



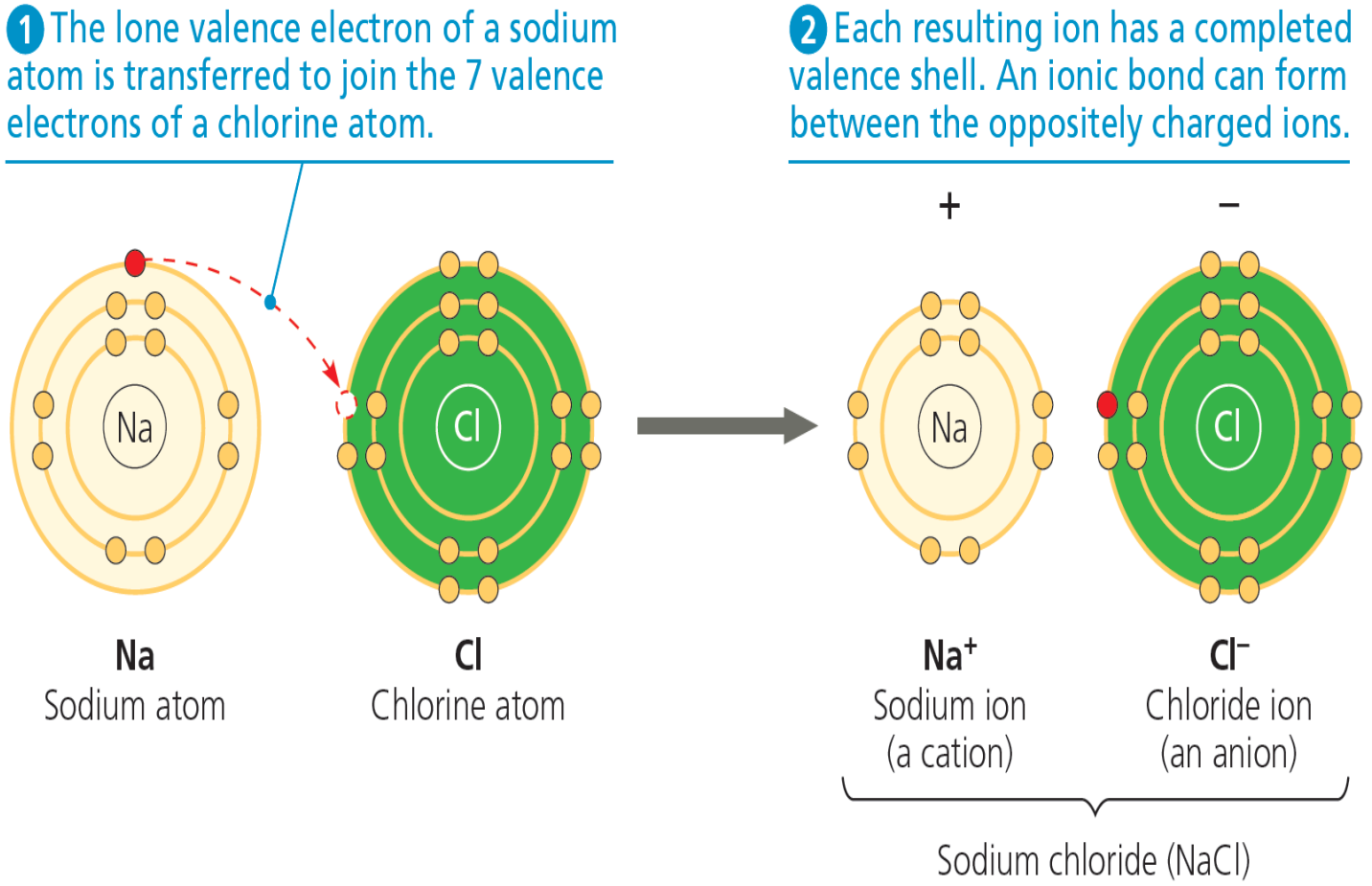
Name and Molecular Formula	Electron Distribution Diagram	Lewis Dot Structure and Structural Formula	Space-Filling Model
(a) Hydrogen (H_2). Two hydrogen atoms share one pair of electrons, forming a single bond.	H H	H:H H—H	
(b) Oxygen (O_2). Two oxygen atoms share two pairs of electrons, forming a double bond.	O O	Ö::Ö O=O	
(c) Water (H_2O). Two hydrogen atoms and one oxygen atom are joined by single bonds, forming a molecule of water.	O H H	:Ö:H H O—H | H	
(d) Methane (CH_4). Four hydrogen atoms can satisfy the valence of one carbon atom, forming methane.	H H C H H	H H:C:H H H | H—C—H | H	

Ionic bonds

- An **ionic bond** can form *if two atoms are so unequal in their attraction for valence electrons that one atom strips an electron completely from the other*.

- After the transfer, both atoms are no longer neutral but have charges and are called **ions**.

- Atoms with positive charges are **cations** and those with negative charges are **anions**.



▲ **Figure 2.12 Electron transfer and ionic bonding.** The attraction between oppositely charged atoms, or ions, is an ionic bond. An ionic bond can form between any two oppositely charged ions, even if they have not been formed by transfer of an electron from one to the other.

Ionic bonds

- Compounds formed by ionic bonds are **ionic compounds**, or **salts**. *An example is NaCl, or table salt.*
  - ✓ The formula for an ionic compound indicates the ratio of elements in a crystal of that salt. NaCl is not a molecule but a salt crystal with equal numbers of Na^+ and Cl^- ions.
- Ionic compounds *can have ratios of elements different from 1:1*.
  - ✓ For example, the ionic compound magnesium chloride ($MgCl_2$) has two chloride atoms for each magnesium atom.
  - ✓ Magnesium needs to lose 2 electrons to drop to a full outer shell; each chlorine atom needs to gain 1 electron.
- Entire molecules that have full electrical charges are also called *ions*.
  - ✓ In the salt ammonium chloride (NH_4Cl), the anion is chloride (Cl^-) and the cation is ammonium (NH_4^+).
- **The strength of ionic bonds depends on environmental conditions**, such as moisture.
- Water can dissolve salts by reducing the attraction between the salt's anions and cations.

Weak bonds: Non-covalent bonds

- Within a cell, **weak, brief bonds** between molecules are important to a variety of processes.

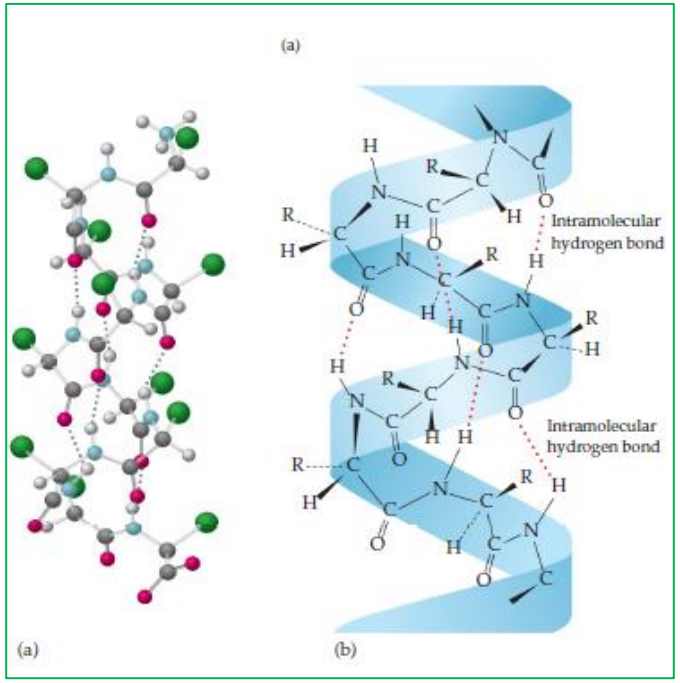
  - ✓ *Many large biological molecules are held in their functional form by weak bonds.*
  - ✓ *When two molecules in the cell make contact, they may adhere temporarily by weak bonds.*

- The **reversibility of weak bonding can be an advantage**: Two molecules can come together, respond to each other in some way, and then separate.

- Weak bonding within and between molecules contributes greatly to the emergent properties of life.

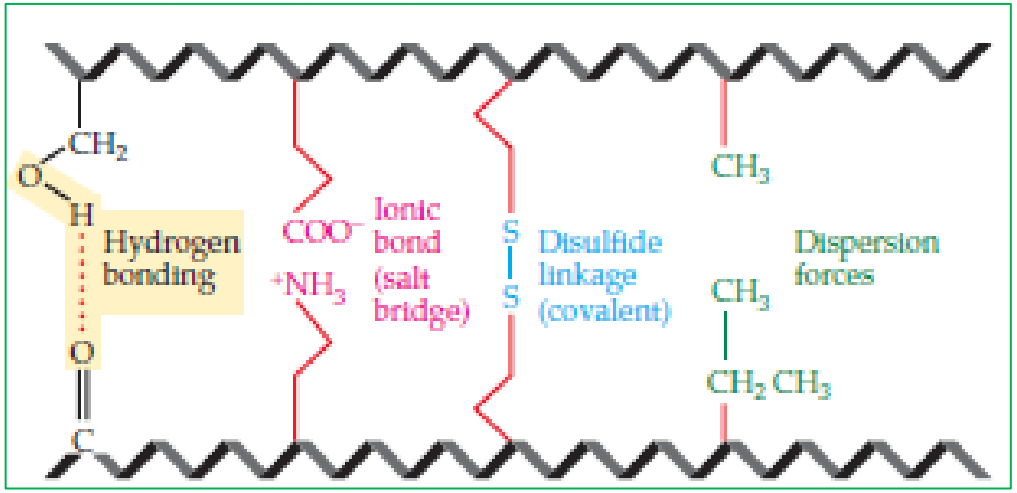
- **Weak interactions** include **ionic bonds between ions dissociated in water**, **hydrogen bonds**, and **van der Waals interactions**.

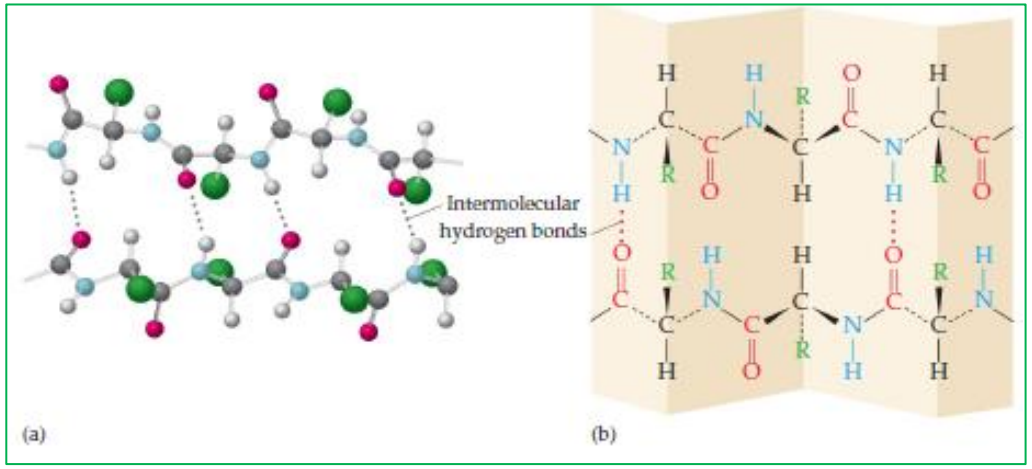
Examples of weak bonds



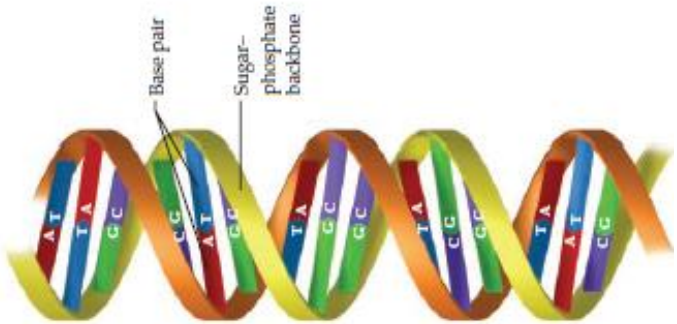
- ✓ *Beta pleated-sheet structure consisting of protein chains.*
- ✓ *Ball-and-stick model.*
- ✓ *Model emphasizing the pleats. The side chains extend above or below the sheet and alternate along the chain. The protein chains are held together by interchain hydrogen bonds.*

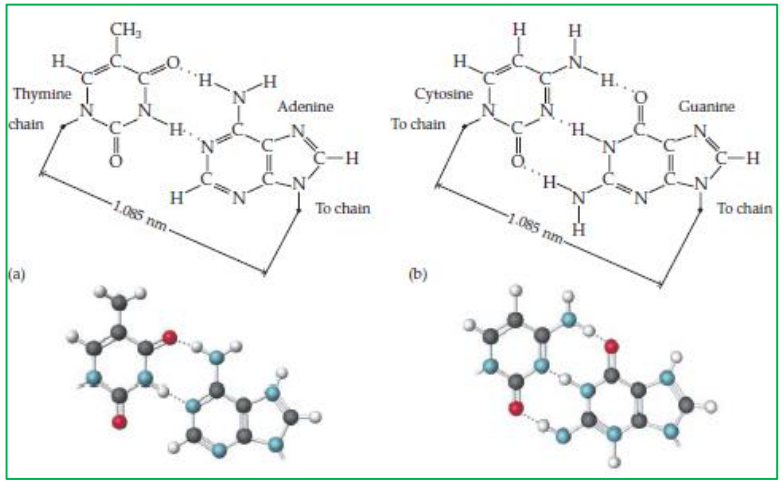
- ✓ *The tertiary structure of proteins is maintained by four different types of forces*





- ✓ *Two representations of the a-helical conformation of a protein chain. (a) Intrachain hydrogen bonding between turns of the helix is shown in the ball-and-stick model. (b) The skeletal representation better shows the helix.*

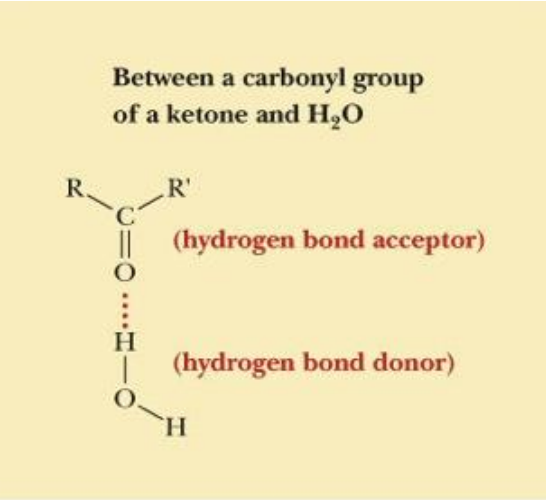


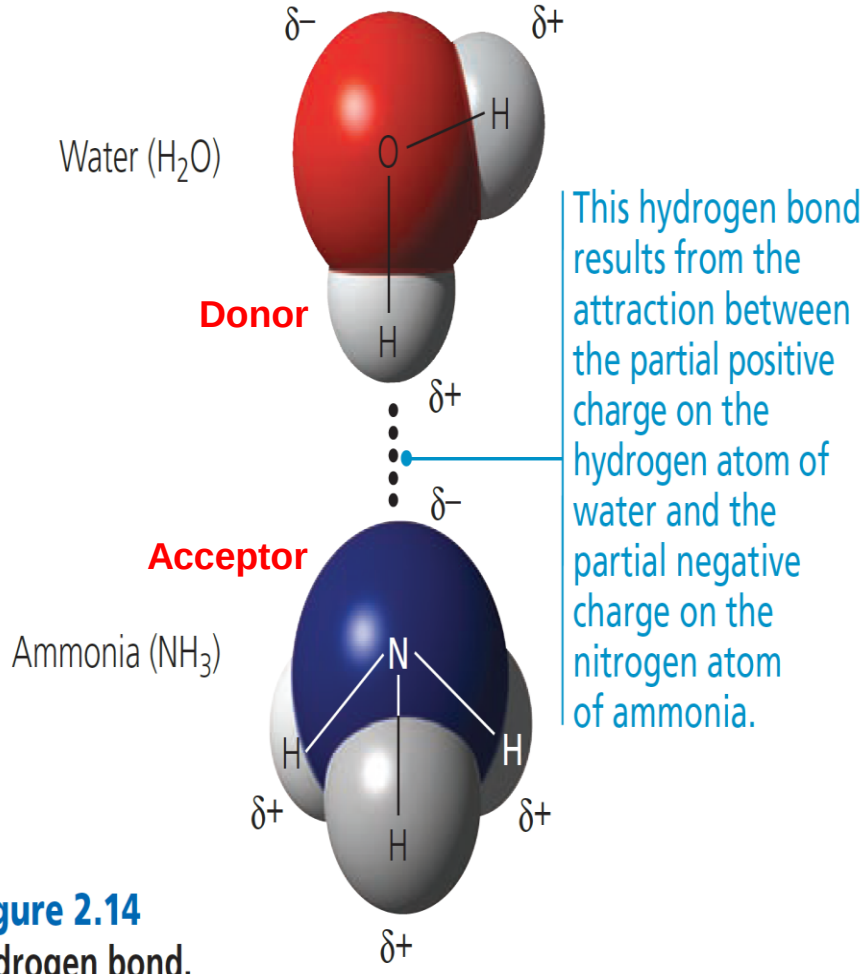


- ✓ *Pairing of the complementary bases*
- ✓ *thymine and adenine (a) and cytosine and guanine (b). The pairing involves*
- ✓ *hydrogen bonding.*

Hydrogen bonds

- **Hydrogen bonds** form when a hydrogen atom that is already covalently bonded to one electronegative atom is attracted to another electronegative atom.
- Hydrogen bonds are denoted as D-H---A
- This indicates that the hydrogen atom is covalently bound to the atom D and favourably interacts with the atom A.
- The atom D is called proton donor, whereas the atom A is called proton acceptor.
- The donor and the acceptor must be electronegative, i.e. hydrogen bonds occur when two electronegative atoms compete for the same hydrogen atom.





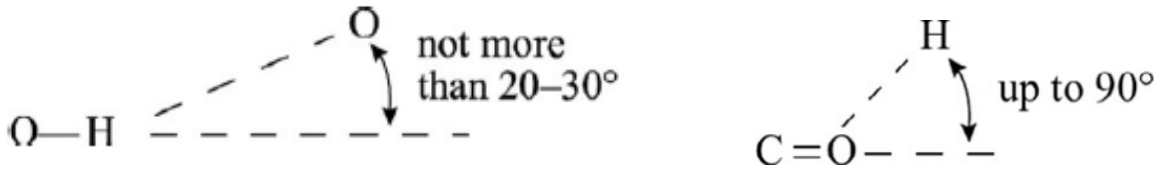
▲ **Figure 2.14**
A hydrogen bond.

Properties

- Hydrogen bonding is determined by the properties of the proton donor and the proton acceptor.

- Unlike van der Waals interactions, H-bonding is rather orientation-sensitive, especially as concerns the orientation of the donor group.

- The distances between the atoms forming the triad D-H"-A are shorter than that expected for van der Waals interactions and in some cases is even close to the corresponding covalent bond.



- Usually, a valence bond of the donor is directed at the acceptor atom (O or N) to be involved in the hydrogen bond.

- It seems that this effect is due to repulsion between O atoms and a very small radius of H atom, when H is in the O–H (or N–H) group.

- Orientation of the acceptor group is of considerably less importance for the H-bond.

Chemical reactions: Make and break bonds

- In **chemical reactions**, chemical bonds are broken and reformed, leading to new arrangements of atoms.

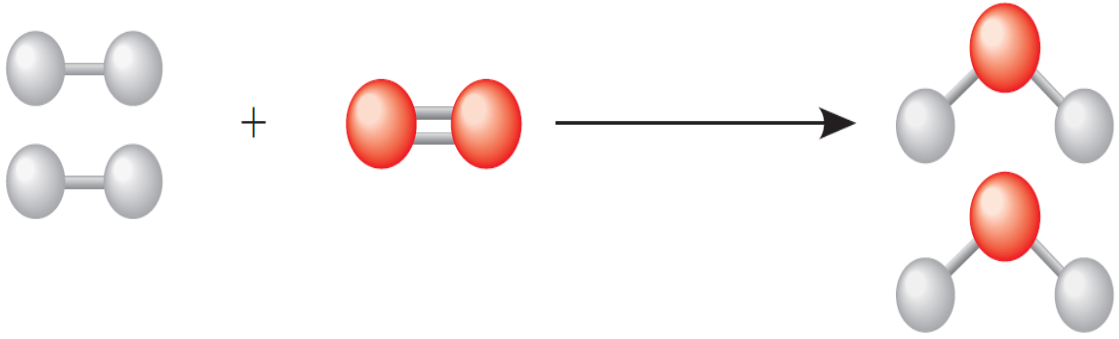
- The starting molecules in the process are called **reactants**, and the final molecules are called **products**.

- In a chemical reaction, all of the atoms in the reactants must be present in the products.

  - ✓ *The reactions must be "balanced."*
  - ✓ *Matter is conserved in a chemical reaction. Chemical reactions rearrange matter; they do not create or destroy matter.*
  - ✓ *For example, the covalent bonds of H_2 and O_2 can recombine to form the new bonds of H_2O. Two molecules of H_2 combine with one molecule of O_2 to form two molecules of H_2O.*

Chemical reactions make and break chemical bonds

The making and breaking of chemical bonds, leading to changes in the composition of matter, are called **chemical reactions**. An example is the reaction between hydrogen and oxygen molecules that forms water:



$2\ H_2 + O_2 \longrightarrow 2\ H_2O$

Reactants **Reaction** **Products**

Some of the types of reactions

- **Photosynthesis** is an important chemical reaction.
  - ✓ *Humans and other animals ultimately depend on photosynthesis for food and oxygen.*
  - ✓ *Powered by sunlight, green plants combine carbon dioxide (CO_2) from the air and water (H_2O) from the soil to create sugar molecules, and they release molecular oxygen (O_2) as a by-product.*
  - ✓ *The overall process of photosynthesis is $6\ CO_2 + 6\ H_2O \rightarrow C_6H_{12}O_6 + 6\ O_2$. (Balance this rxn: $CO_2 + H_2O \rightarrow C_6H_{12}O_6 + O_2$)*
  - ✓ *This process occurs in a sequence of individual chemical reactions that rearrange the atoms of the reactants to form the products.*

- **All chemical reactions are reversible**, with the products in the forward reaction becoming the reactants in the reverse reaction.
  - ✓ *For example, in the reaction $3\ H_2 + N_2 \leftrightarrow 2\ NH_3$, hydrogen and nitrogen molecules combine to form ammonia, but ammonia can decompose to hydrogen and nitrogen molecules.*

- Eventually, the rate of formation of products is the same as the rate of breakdown of products (formation of reactants), and the system is at **chemical equilibrium**.
  - ✓ *At equilibrium, products and reactants are continually being formed, but there is no net change in the concentrations of reactants and products.*
  - ✓ *At equilibrium, the concentrations of reactants and products are typically not equal, but their concentrations have stabilized at a particular ratio.*

Chemical equilibrium

- Consider the hypothetical reaction below where *a* moles of a compound A react with *b* moles of a compound B to yield *c* moles of a compound C and *d* moles of a compound D, as illustrated:

$$aA + bB \rightleftharpoons cC + dD$$

- The equilibrium constant for this reaction is given by:

$$K = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

- Even though *K* is a constant, the equilibrium composition for a reaction does not need to be the same. For example, take where the values of *a*, *b*, *c* and *d* are all 1, that is:

$$A + B \rightleftharpoons C + D$$

- If this reaction had an equilibrium constant of *K* = 1, then clearly if [A] = [B] = [C] = [D] = 1 mol L^{-1}, the system would be at equilibrium, because

$$K = \frac{[C][D]}{[A][B]} = \frac{1 \times 1}{1 \times 1} = 1$$

- However, the value of *K* would also be 1 if the composition was [A] = 10 mol L^{-1}, [B] = 0.1 mol L^{-1}, [C] = 50 mol L^{-1} and [D] = mol L^{-1}, thus

$$K = \frac{[C][D]}{[A][B]} = \frac{50 \times 0.02}{10 \times 0.1} = 1$$

  - *In summary, the above calculation tells us both mixtures would be at equilibrium! The equilibrium constant depends on the initial amounts of material.*

Units

Metric prefix	Abbreviation	Power of 10
giga-	G	10^{9}
mega-	M	10^{6}
kilo-	k	10^{3}
milli-	m	10^{-3}
micro-	μ	10^{-6}
nano-	n	10^{-9}
pico-	p	10^{-12}
femto-	f	10^{-15}
atto-	a	10^{-18}

Units

LENGTH		
1 km (kilometer)	=	10^3 m
1 m (meter)		
1 cm (centimeter)	=	10^{-2} m
1 mm (millimeter)	=	10^{-3} m
1 μm (micrometer)	=	10^{-6} m
1 nm (nanometer)	=	10^{-9} m
1 Å (Ångström)	=	10^{-10} m
MASS		
1 kg (kilogram)	=	10^3 g
1 g (gram)		
1 mg (milligram)	=	10^{-3} g
1 μg (microgram)	=	10^{-6} g
1 ng (nanogram)	=	10^{-9} g

VOLUME					
1 l (liter)	=		=	$(10^{-1}\ m)^3$	
1 ml (milliliter)	=	10^{-3} l	=	$(10^{-2}\ m)^3$	= 1 cm^3
1 μl (microliter)	=	10^{-6} l	=	$(10^{-3}\ m)^3$	= 1 mm^3
1 nl (nanoliter)	=	10^{-9} l	=	$(10^{-4}\ m)^3$	
CONCENTRATION					
1 M (molar)	=	1 mole/l = 6.02×10^{23} molecules/l			
1 mM (millimolar)	=	10^{-3} M			
1 μM (micromolar)	=	10^{-6} M			
1 nM (nanomolar)	=	10^{-9} M			
USEFUL CONSTANTS, CONVERSIONS, AND DEFINITIONS					
1 mole	=	6.02×10^{23} molecules			
1 c (calorie)	=	heat needed to raise the temperature of 1 g of water by 1°C			
1 J (joule)	=	0.239 c			
1 kcal (kilocalorie)	=	10^3 c = 4.18 kJ (kilojoules)			
1 l water	=	1 kg (at 4° C)			
1 d (dalton)	=	approximate mass of one hydrogen atom (1.7×10^{-24} g)			
1 kd (kilodalton)	=	10^3 d			
Mass of Earth	=	10^{24} kg			
Bacterial genomes	=	$0.5–5 \times 10^6$ nucleotide pairs, depending on organism			
Human genome	=	3×10^9 nucleotide pairs (haploid)			